

Global Architectural Glass Building Codes Comparison

US IBC | EU EN | China GB | Australia AS

A practical reference for architects, engineers, and
procurement managers working with architectural glass
across international markets.

IBC / ASTM
United States

EN 12150 / 14449
European Union

GB 15763 / 9962
China

AS 1288 / 2208
Australia

SINCERE GLASS

sincerglass.com

Architectural Glass Manufacturer | Wuhan, China | Est. 2005

2026 Edition

Table of Contents

- 1. Introduction & How to Use This Guide
- 2. Tempered (Toughened) Glass Standards
- 3. Laminated Glass Standards
- 4. Insulated Glass Unit (IGU) Standards
- 5. Low-E & Coated Glass Standards
- 6. Safety Glazing Requirements by Application
- 7. Testing & Certification Requirements
- 8. Quick Reference: Standards Cross-Map
- 9. About Sincere Glass

1. Introduction

Navigating architectural glass regulations across international markets is one of the most complex challenges facing architects, engineers, and procurement teams today. Each region maintains its own framework of standards governing safety, thermal performance, and structural integrity — and non-compliance can mean project delays, costly redesigns, or outright rejection at inspection.

This guide provides a practical, side-by-side comparison of the four major regulatory systems that govern architectural glass worldwide:

- **United States** — IBC (International Building Code) + ASTM standards
- **European Union** — EN harmonized standards under the CPR
- **China** — GB mandatory national standards
- **Australia** — AS/NZS standards referenced by the NCC

How to Use This Guide

Each section covers one glass type, with a comparison table showing the key standard numbers, requirements, and test methods across all four regions. Use the Quick Reference cross-map in Section 8 to find equivalents when specifying glass for multi-market projects.

Note: Standards are current as of early 2026. Always verify with the latest published edition before specifying. This document is for reference only and does not constitute engineering or legal advice.

2. Tempered (Toughened) Glass Standards

Tempered glass is heat-treated to create surface compression, making it 4–5× stronger than annealed glass. When broken, it fragments into small, relatively harmless pieces. All four regions classify it as a safety glazing material, but specific requirements differ.

2.1 Primary Standards Comparison

Requirement	US (ASTM)	EU (EN)	China (GB)	Australia (AS)
Primary Standard	ASTM C1048 Standard Specification for Heat-Strengthened and Fully Tempered Flat Glass	EN 12150-1 Thermally toughened soda lime silicate safety glass	GB 15763.2 Safety glazing materials — Part 2: Tempered glass	AS/NZS 2208 Safety glazing materials in buildings
Fragmentation Test	Min. 40 particles in 50×50 mm area (for thickness ≤3.2 mm: 30 particles)	Min. 40 particles in 50×50 mm area; max. single fragment ≤65 mm for ≤3 mm thickness	Min. 40 particles in 50×50 mm area; no single particle >75 mm length	Min. 40 particles in 50×50 mm area (references EN 12150-1 test method)
Surface Compression	≥69 MPa (10,000 psi) for fully tempered	≥75 MPa (edge compression); surface not directly specified but implied by frag test	≥69 MPa surface compressive stress	Not directly specified; validated through fragmentation test
Bow / Roller Wave	Overall bow ≤3 mm/m; local bow ≤0.5 mm/300 mm	Overall bow: 3 mm/m; local bow: 0.5 mm/300 mm (EN 12150-1 Table 4)	Overall bow ≤3 mm/m	Bow limits in AS 2208 Appendix
Thickness Range	3–19 mm typical; thinner by special order	3–19 mm (EN 12150-1 covers 3–25 mm)	4–19 mm per GB 15763.2	3–19 mm per AS 2208
Marking	Permanent label or stamp; SGCC certification mark if certified	CE marking mandatory; EN 12150 + notified body number	CCC (China Compulsory Certification) mark required	AS/NZS 2208 compliance mark; SAI Global or JAS-ANZ certified

2.2 Key Differences to Watch

- **Certification:** The US uses voluntary SGCC certification, while China requires mandatory CCC and the EU requires CE marking with Declaration of Performance (DoP).
- **Edge quality:** EN 12150-1 has explicit edge quality classes (arris, smooth ground, polished) tied to intended use; ASTM C1048 defers to project specifications.
- **Heat Soak Test:** EN 14179-1 provides a standardized HST protocol to reduce spontaneous breakage risk from nickel sulfide (NiS) inclusions. ASTM has no direct equivalent standard, though ASTM C1866 covers heat-soaked tempered glass. China GB/T 33000 addresses HST.

3. Laminated Glass Standards

Laminated glass consists of two or more glass plies bonded with an interlayer (typically PVB or SGP). It provides post-breakage retention, sound insulation, and UV filtering. Structural and security applications demand specific interlayer performance.

3.1 Primary Standards Comparison

Requirement	US (ASTM)	EU (EN)	China (GB)	Australia (AS)
Primary Standard	ASTM C1172 Specification for Laminated Architectural Flat Glass	EN 14449 Laminated glass and laminated safety glass	GB 15763.3 Safety glazing — Part 3: Laminated glass	AS/NZS 2208 + AS 1288 Section 7
Impact Classification	CPSC 16 CFR 1201 Category I (≤400 ft-lb) Category II (≤800 ft-lb)	EN 12600 Pendulum test classes: 1B1 / 2B2 / 3B3 (drop heights 1200/450/190 mm)	GB 15763.3 Impact test per 6.5: 1219 mm drop height (45 kg shot bag)	AS/NZS 2208 References EN 12600 pendulum test; Classification A, B, C
Interlayer Requirements	PVB: ASTM C1651 Ionoplast: manufacturer specs Min. 0.38 mm PVB typical	EN ISO 12543-2 Interlayer performance; PVB per EN 12543-4	PVB film per JC/T 637 or GB/T 31832; Min. 0.38 mm	Per AS 2208; Interlayer conforming to manufacturer declaration
Acoustic Performance	ASTM E90 / E413 STC (Sound Transmission Class) No mandatory minimum	EN ISO 10140 test method; EN 717-1 (R_w rating) Declared in DoP	GB/T 8485 for sound insulation measurement; Required for curtain walls (JGJ 102)	Tested per AS 1191; $R_w + C_{tr}$ rating; BCA Section J energy provisions
Security / Bullet Resistance	UL 752 (Bullet Resistant) ASTM F1233 (Forced Entry) Levels 1–8 + special	EN 1063 (Bullet Resistant) EN 356 (Manual Attack) Classes BR1–BR7 / P1A–P8B	GA 165 (Bullet Resistant) GB 17840 (Bank security) Classes FB1–FB7	AS/NZS 2343 (Bullet Resistant) Levels G0–G2, R1–R3, S
UV Transmittance	PVB blocks >99% UV; no mandatory standard	EN 410 for calculation; PVB typically blocks >99%	GB/T 2680 for UV transmittance measurement	AS 2047 / WERS system; UV not separately rated

3.2 Key Differences to Watch

- **Impact classification systems are not interchangeable.** CPSC Category II ≠ EN 12600 Class 1B1, even though both represent the highest safety tier. Cross-referencing requires testing to each standard separately.
- **Overhead glazing:** All regions require laminated glass for overhead applications, but minimum interlayer thickness and retention requirements vary. IBC Section 2409, EN 12600 + national annexes, GB 50009, and AS 1288 Section 7 each set different rules.
- **Structural interlayers:** SGP (SentryGlas) and similar ionoplast interlayers have 5× the stiffness of PVB. EU and China increasingly reference specific interlayer performance requirements for structural glass applications.

4. Insulated Glass Unit (IGU) Standards

Insulated glass units (IGUs) consist of two or more glass panes separated by a sealed air or gas-filled cavity. They are the primary means of achieving thermal and acoustic insulation in building envelopes. Standards cover both the unit's construction and its thermal performance.

4.1 Primary Standards Comparison

Requirement	US (ASTM)	EU (EN)	China (GB)	Australia (AS)
Primary Standard	ASTM E2190 Insulating Glass Unit Performance & Evaluation	EN 1279 (Parts 1–6) Insulating glass units	GB/T 11944 Insulating glass + JGJ 102 (curtain wall code)	AS 4666 Insulating glass units + AS 2047 (windows)
Seal Durability (Moisture Penetration)	ASTM E2190: Fog test + dew point test; Initial dew point $\leq -40^{\circ}\text{C}$	EN 1279-2: Moisture penetration index (I); EN 1279-3: gas leakage rate	GB/T 11944: Initial dew point $\leq -40^{\circ}\text{C}$; Durability per Annex A	AS 4666: Initial dew point $\leq -40^{\circ}\text{C}$; Accelerated weathering test
Thermal Performance	NFRC 100 (U-factor) NFRC 200 (SHGC) ASTM C1199 / E1423	EN 673 (U-value calc) EN 410 (solar/light) EN ISO 10077-1 (windows)	GB/T 2680 (solar/light) GB/T 22476 (U-value calc) GB 50176 (building thermal)	WERS (Window Energy Rating Scheme); AS 2047 testing protocol
Gas Fill Requirements	No mandatory minimum; Argon fill measured per ASTM E2649 ($\geq 90\%$ initial)	EN 1279-3: Gas concentration $\geq 90\%$; Leakage rate $\leq 1\%$ /year	GB/T 11944: Argon $\geq 90\%$ initial fill; Annual leakage test	AS 4666: Gas fill per manufacturer declaration; $\geq 90\%$ typical
Spacer Requirements	Aluminum, stainless steel, or warm-edge (foam/hybrid); No mandatory warm-edge	EN 1279-1: Spacers tested as system; Warm-edge TPS/flexible common	Aluminum or warm-edge; Warm-edge encouraged per GB 50189 (energy code)	Per AS 4666; No mandatory spacer type; Warm-edge for high-performance
Max Size	No standard limit; Manufacturer capability (Sincere Glass: 3m $\times$ 15m)	No standard limit; EN 1279 applies to all sizes	No standard limit; JGJ 102 governs structural calculation for large panels	Per manufacturer; Design per AS 1170 wind load calculations

4.2 Key Differences to Watch

- **Performance rating systems differ significantly.** The US NFRC system rates whole-window U-factor (including frame), while EN 673 rates center-of-glass U-value. Direct numeric comparison is misleading without noting the boundary conditions.
- **Gas retention testing:** EN 1279-3 uniquely specifies an ongoing gas leakage rate ($\leq 1\%$ /year), providing a long-term performance guarantee not explicitly required by ASTM E2190.
- **Altitude/pressure compensation:** IGUs shipped to high-altitude sites need capillary tubes or pressure equalization. China's JGJ 102 has specific provisions for this due to the wide altitude range across Chinese provinces.

5. Low-E & Coated Glass Standards

Low-emissivity (Low-E) coatings reduce heat transfer through glass by reflecting infrared radiation. Two main types exist: hard coat (pyrolytic, applied during float process) and soft coat (sputtered, applied in a vacuum chamber). Regulatory focus is on energy performance rather than the coating process itself.

5.1 Primary Standards Comparison

Requirement	US (ASTM)	EU (EN)	China (GB)	Australia (AS)
Coating Standard	ASTM C1376 Pyrolytic & vacuum-deposited coatings on flat glass	EN 1096 (Parts 1–4) Coated glass	GB/T 18915.1 (soft coat) GB/T 18915.2 (hard coat)	No specific coating std; Performance tested per AS 2047 / WERS
Emissivity Requirements	NFRC uses emissivity in U-factor calculation; Typical ≤0.15 (soft coat)	EN 12898 for measurement; Declared in CE DoP; ≤0.10 common for triple-silver	GB/T 2680 measurement; GB 50189 sets max U-value implying emissivity targets	WERS declaration; No mandatory emissivity value; performance-based
Energy Code Compliance	IECC / ASHRAE 90.1 Prescriptive U + SHGC by climate zone (1–8)	EPBD + national codes (e.g., EnEV, RT 2020, Part L) U-value + g-value limits	GB 50189 + GB 55015 (2021 update) Prescriptive by climate zone	NCC Section J (Vol. 1) Total System U-value + SHGC by climate zone
Durability Testing	ASTM C1376: Abrasion, humidity, UV, salt spray resistance	EN 1096-2: Climate cycle, SO ₂ , acid, abrasion, UV tests (Class A/B/C/D)	GB/T 18915: Salt spray, UV, humidity, abrasion, acid resistance	Per manufacturer; NATAs-accredited lab testing recommended
Surface Position	NFRC designates surfaces 1 (exterior) to 4; Low-E typically on #2 or #3	Same surface numbering; EN 410 specifies calculation by position	Same numbering system; GB/T 2680 calculation by surface position	Same numbering system; WERS calculation by surface position

5.2 Key Differences to Watch

- **Climate zones drive everything.** A Low-E coating optimized for hot climates (low SHGC, surface #2) may be wrong for cold climates (high SHGC, surface #3). Each region's energy code has its own climate zone map.
- **Soft coat handling:** Sputtered Low-E must be within an IGU cavity (not exposed). All four standards agree on this, but field installation practices vary widely.
- **Self-cleaning coatings:** EN 1096-5 specifically covers self-cleaning coated glass; other regions handle this under general coating durability standards.

6. Safety Glazing Requirements by Application

Beyond material standards, building codes specify which glass types are permitted (or required) for specific applications. This section maps the most common applications across all four systems.

6.1 Application Requirements

Application	US (IBC)	EU (EN / national)	China (GB / JGJ)	Australia (NCC)
Overhead Glazing	IBC 2409: Laminated required; Fully tempered only if screened below	National annexes vary; Laminated or lam+tempered; EN 12600 Class 1	JGJ 102 Section 4.4: Laminated mandatory; PVB ≥1.14 mm for overhead	AS 1288 Section 7: Grade A safety glass; Laminated required overhead
Doors & Sidelights	IBC 2406: Safety glazing to CPSC 16 CFR 1201; Top to bottom	EN 12600 Class 1 or 2; Fully tempered or lam; Per national building code	GB 50009 / JGJ 113: Tempered or laminated; School doors: laminated	AS 1288 Section 6: Grade A safety glass to AS 2208
Balustrades & Railings	IBC 2407: Laminated or tempered+lam; Guard loading 890 N/m	EN 12600 Class 1(1)1; Structural calculation per national Eurocode annex	JGJ 113: Tempered laminated; ≥2 plies for structural	AS 1288 Section 8: Grade A; balustrade calc per AS/NZS 1170
Wet Areas (Bathrooms)	IBC 2406: Safety glazing if within 457 mm of tub/shower	Per national building regs; EN 12600 safety class typically required	GB 50210: Tempered or laminated in wet areas	AS 1288: Grade A within 2 m of shower/bath
Curtain Walls	IBC 2403 + ASCE 7: Wind load design; No mandatory glass type	EN 13830 + Eurocode 1: Wind load design; EN 14019 air/water/wind test	JGJ 102: Comprehensive curtain wall code; mandatory for high-rise (>28m)	AS 2047 + AS/NZS 1170: Wind load design; Verification per AS 1288
Fire-Rated Glazing	ASTM E119 / UL 263: Integrity (E) + insulation (I); 20/45/60/90/120 min ratings	EN 13501-2: E / EW / EI classification; 15/20/30/60/90/120 min	GB 15763.1: Fire-resistant glass Class A (EI) / Class B (E)	AS 1530.4: -/30/60/90/120/180 Integrity + insulation

7. Testing & Certification Requirements

Understanding certification pathways is critical for international glass trade. A product certified for one market often needs additional testing and documentation for another.

7.1 Certification Systems

Aspect	United States	European Union	China	Australia
Mandatory Certification	CPSC 16 CFR 1201 (safety glazing); Voluntary: SGCC, IGCC	CE marking mandatory under CPR for all construction products	CCC mandatory for safety glass; Voluntary: CQC mark	WaterMark (plumbing) CodeMark (voluntary); AS compliance required
Testing Bodies	Accredited labs (e.g., ICC-ES, UL, Intertek); NVLAP accreditation	Notified Bodies per Member State; EN ISO 17025 labs	CNCA-designated labs; CCC testing centers; CNAS-accredited labs	NATA-accredited labs; JAS-ANZ certified bodies
Factory Audit	SGCC/IGCC: annual factory + unannounced audits; CPSC: none	CE: Factory Production Control (FPC) per EN 14351 / EN 13830	CCC: initial factory inspection + annual follow-up audits	CodeMark: scheme rules define audit; Product certification
Documentation	Test reports; Certificate of compliance; shop drawings	Declaration of Performance (DoP); CE label; FPC manual	CCC certificate; Test report; Product data sheet	Test report; Certificate of conformity; compliance statement
Timeline (Typical)	4–8 weeks for SGCC; 2–4 weeks for lab testing	6–12 weeks for full CE; 3–6 weeks lab testing	8–16 weeks for CCC; 4–8 weeks lab testing	4–8 weeks lab testing; CodeMark: 8–12 weeks

Pro tip: When sourcing glass from China for export, request that the manufacturer hold both CCC (for domestic compliance) and test reports to the destination market's standards. Dual-certified factories can significantly reduce your project timeline.

8. Quick Reference: Standards Cross-Map

Use this table to quickly find the equivalent standard when converting specifications between markets.

Glass Type / Topic	US	EU	China	Australia
Tempered glass	ASTM C1048	EN 12150-1	GB 15763.2	AS/NZS 2208
Heat-soaked tempered	ASTM C1866	EN 14179-1	GB/T 33000	AS/NZS 2208 App.
Laminated glass	ASTM C1172	EN 14449	GB 15763.3	AS/NZS 2208
Insulated glass units	ASTM E2190	EN 1279	GB/T 11944	AS 4666
Low-E coating	ASTM C1376	EN 1096	GB/T 18915	N/A (perf. based)
Safety glazing impact	CPSC 16 CFR 1201	EN 12600	GB 15763	AS/NZS 2208
Solar/optical props.	NFRC 200	EN 410	GB/T 2680	WERS
Thermal (U-value)	NFRC 100	EN 673	GB/T 22476	WERS / AS 2047
Curtain wall design	ASCE 7 + IBC	EN 13830	JGJ 102	AS 2047
Fire-rated glass	ASTM E119	EN 13501-2	GB 15763.1	AS 1530.4
Bullet-resistant	UL 752	EN 1063	GA 165	AS/NZS 2343
Building energy code	IECC / ASHRAE 90.1	EPBD	GB 50189	NCC Section J

This cross-map covers the most commonly referenced standards. Some regions have additional state/provincial or project-specific requirements. Always check with local authorities having jurisdiction (AHJ).

9. About Sincere Glass

Sincere Glass is an architectural glass manufacturer based in Wuhan, China, with over 15 years of experience in the domestic Chinese market. Our 20,000 m² facility houses a complete production line for tempered, insulated, laminated, enameled, and Low-E glass.

Production Capabilities

- **Jumbo Size Processing:** Our tempering furnaces and laminating autoclaves handle panels up to **3m × 15m** — among the largest in central China.
- **4 Intelligent Cutting Lines** with CNC optimization for minimal waste.
- **4 Edge-Grinding Lines** for precision edge work across all glass types.
- **2 Tempering Furnaces** with convection heating for optical-quality flat tempered glass.
- **2 Insulated Glass Lines** including an ultra-large automatic argon gas-fill line.
- **2 High-Pressure Autoclaves** for laminated and structural glass up to 3m × 15m.

Why Work With Us

We combine mid-scale production flexibility with large-factory capability. Whether your project needs standard tempered panels or jumbo-format laminated curtain wall units, our engineering team can work with your specifications and help navigate compliance requirements for your target market.

Get in Touch

Website	sincereglass.com
Email	info@sincereglass.com
Location	Wuhan, Hubei Province, China
Capabilities	Tempered • Insulated • Laminated • Enameled • Low-E • Curved

Request a quote or ask our engineering team about glass specifications for your project at sincereglass.com/contact